# FlyCommerce OS: Capstone Impact Project Summary

8-Week Intensive Product Design Sprint · Track: UX/UI Design · Candidate: K. M. Siam

**24H CAPSTONE IMPACT PACKAGE**

| CANDIDATE | CAPSTONE FOCUS | PRIMARY TOOLING | TOTAL VERIFIED HOURS |
|---|---|---|---|
| K. M. Siam (@KMSiam) | Omnichannel Inventory Hub | Figma, Claude 3.5, DeepSeek | 40.0 Program Hours Logged |

| SYSTEM USABILITY SCORE | TIME ON TASK | TASK COMPLETION | WCAG 2.1 AA/AAA |
|---|---|---|---|
| **89.0** (+17.5) | **54 sec** (-72%) | **96.7%** | **100% Pass** |
| Top 5% percentile (Grade A+ Excellent) | Stock triage dropped from 3m 12s | Tested across 5 enterprise operations leads | Zero detached tokens; 7.3:1 text contrast |

## 1. Project Snapshot (Honest Reality for Incoming Designers)

**Core Problem:** Multi-channel DTC brands scaling on Shopify, TikTok Shop, and Amazon FBA suffer a 42-minute batch inventory sync lag. Viral demand spikes deplete stock on one channel while others oversell phantom units, causing $180,000+ in annual fines and stockouts.
**What We Built:** An AI-native inventory operating system combining high-density real-time data grids with an explainable AI stock rebalancing governor (60s undo window + dynamic safety interlock) and 54px touch-friendly warehouse tablet modes.

## 2. Week-by-Week Journey Map (Deliverables & AI Integration)

| Week | Code | Key Shipped Deliverable | AI Tools Used | Role of AI |
|---|---|---|---|---|
| Wk 1 | UX-01 | Capstone Brief & AI Design Toolkit | Claude 3.5, Figma AI | Problem scoping & workflow setup |
| Wk 2 | UX-02 | Persona, Journey Map & 5 Interviews | Claude 3.5, ChatGPT-4o | Qualitative thematic interview synthesis |
| Wk 3 | UX-03 | Information Architecture & User Flows | Claude 3.5 (Mermaid) | Algorithmic state branching & flow maps |
| Wk 4 | UX-04 | Kill Your Darlings (3D Pivot) & Wireframes | First Draft, Relume | Comparative low-fidelity wireframing |
| Wk 5 | UX-05 | FlyCore Design System: Tokens & Variables | Claude 3.5, Tokens Studio | Semantic variable token architecture |
| Wk 6 | UX-06/08 | 15+ Hi-Fi Screens, Variants & Prototype | DeepSeek-R1, Recraft.ai | 36-variant matrix & 10 edge cases |
| Wk 7 | UX-07/09 | AI Surface (HAX), Usability (5 Ops) & WCAG | Claude 3.5, WebAIM, Stark | HAX trust audit, testing & token fixes |
| Wk 8 | UX-11/12 | NN/g Case Study & Dev Hand-off Package | Claude 3.5, Playwright | Full portfolio report & 40h verified log |

## 3. AI Integration Log (Top 3 Prompts & 1 Failed Prompt Lesson)

### Top 3 Time-Saving Prompts (Saved 13+ Hours)

1. Component Variant Matrix Generator:

> *"Output a full markdown table listing every unique combination of Button [Hierarchy] × [Size] × [State] with exact auto-layout padding, corner radii, and token bindings."*

2. DeepSeek Edge Case Generator:

> *"Brainstorm 10 realistic supply-chain edge cases (API timeout, negative CSV quantities) with plain-English user recovery copy."*

3. Microsoft HAX Compliance Audit:

> *"Audit our AI stock drawer against HAX guidelines G1, G8, G9, G11. Suggest concrete Shape of AI patterns for failures."*

### The 1 Failed Prompt Lesson (Where AI Failed)

Automated WCAG Color Palette Generation:

We prompted LLMs to generate a high-contrast accessible color palette. The model outputted #64748B on white and claimed it passed WCAG AA. In reality, actual contrast checkers measured **4.4:1 (Fail)**, and alert text measured **2.1:1 (Fail)**.

> Lesson: Never trust LLMs with math, color luminance, or physics calculations. Mandatory human verification using WebAIM tools is required. We manually hardened tokens to #475569 (7.3:1 AAA Pass).

## 4. Tangible Assets Shipped for FlyCommerce OS

**1. Master Figma File**
6 structured pages: Tokens, Variables, 15+ Hi-Fi Screens, and Dev Redlines.

**2. Tested Prototype**
3 named user journeys with Smart Animate and dynamic safety interlocks.

**3. NN/g Case Study**
Full public portfolio writeup with AI workflow logs and critical reflections.

## 5. Reflections: Top 3 Learnings & 1 Major Challenge Overcome

### Top 3 Learnings

- **Tokens Over Pixels:** Semantic variables allowed remediating 12 contrast bugs in 20 minutes across 15+ screens.
- **AI is a Synthesizer, Not an Architect:** AI excels at combinatorial matrices, but lacks human restraint and empathy.
- **Enterprise Users Crave Density:** E-commerce operators reject spacious whitespace in favor of monospace tabular data rows.

### Major Challenge Overcome

**Overcoming the AI "Black-Box" Trust Barrier:** Operators initially feared auto-rebalance tools would silently deplete stock. We introduced the **3-Tier AI Governor** (inline formula transparency, physical slider safety locks when reserve <15%, and a 60s reversible undo toast), increasing SUS from 71.5 to 89.0 (Grade A+).

**Master Deliverable Links (View-Only & Logged-Out Access Tested):**
Figma Workspace: Figma Master File · Interactive Prototype: Live Prototype · Case Study: NN/g Case Study · GitHub Repo: Flyrank-Internship-UX-UI-Design